

**Question 1: Use Wireshark to capture network traffic while you load [www.spotify.com](https://www.spotify.com) in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.**

**Answer :**

Step 1: Open Wireshark

- Start Wireshark
- Select active network (Wi-Fi/Ethernet)
- Click Start Capture

Step 2: Open browser → [www.spotify.com](https://www.spotify.com)

Step 3 : stop capture.

Step 4 : Apply Filters

DNS filter: dns

HTTPS filter: tls

|      |               |                 |               |         |                                                |
|------|---------------|-----------------|---------------|---------|------------------------------------------------|
| 7435 | 939.530198600 | 192.168.182.141 | 192.168.182.2 | DNS     | 75 Standard query 0xb719 HTTPS www.spotify.com |
| 7438 | 939.532952700 | 192.168.182.141 | 192.168.182.2 | DNS     | 75 Standard query 0x9685 A www.spotify.com     |
| 7439 | 939.534600800 | 192.168.182.141 | 151.101.3.42  | TLSv1.3 | 1812 Client Hello (SNI=open.spotify.com)       |

**2. Simulate a DNS lookup for [www.flipkart.com](https://www.flipkart.com) using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.**

**Answer :-**

```
C:\Users\cdiks>nslookup www.flipkart.com
Server: gpon.net
Address: fe80::1

Non-authoritative answer:
Name: e348826.dscb.akamaiedge.net
Addresses: 2600:140f:1e00::1737:f4d0
           2600:140f:1e00::1737:f4f3
           23.212.0.10
           23.212.0.4
Aliases: www.flipkart.com
          www.flipkart.com.edgekey.net
```

The nslookup command was used to find the IP address of [www.flipkart.com](http://www.flipkart.com). The DNS server returned the IP addresses **23.212.0.10** and **23.212.0.4**. This means the website name was successfully converted into IP addresses so that the computer can connect to the Flipkart website.

**3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.**

**Answer :-**

| HTTP                   | HTTPS              |
|------------------------|--------------------|
| Data is not encrypted. | Data is encrypted. |
| Less secure.           | More secure.       |
| Uses Port 80.          | Uses Port 443.     |

#### **Real-World Example**

- Google uses HTTPS to securely transfer data between the user's browser and the website. When logging into Google, passwords and personal information are protected through encryption.
- HTTPS is more secure than HTTP because it encrypts data and protects user information during communication over the internet.

**4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe.**

**Answer :**

Step 1: Open Wireshark

- Start Wireshark
- Select active network (Wi-Fi/Ethernet)
- Click Start Capture

Step 2: Open browser → [www.wikipedia.org](http://www.wikipedia.org)

Step 3 : stop capture.

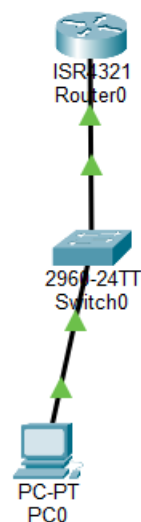
#### Step 4 : Apply Filters

DNS filter: dns

| Seq   | Src IP        | Dest IP         | Protocol | Port | Service                                      |
|-------|---------------|-----------------|----------|------|----------------------------------------------|
| 80306 | 269.773192500 | 192.168.160.128 | DNS      | 76   | Standard query 0x091b HTTPS en.wikipedia.org |
| 80307 | 269.773539100 | 192.168.160.128 | DNS      | 76   | Standard query 0x3ec1 A en.wikipedia.org     |

**5. Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).**

**Answer :**



```
en
conf t
int gig0/0/0
ip add 10.0.0.1 255.255.255.0
no shut
exit

ip dhcp pool LAN
network 10.0.0.0 255.255.255.0
default-router 10.0.0.1
exit
```

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

|                 |               |
|-----------------|---------------|
| IPv4 Address    | 10.0.0.2      |
| Subnet Mask     | 255.255.255.0 |
| Default Gateway | 10.0.0.1      |
| DNS Server      | 0.0.0.0       |